

NetBox™ Hardening Guide

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LenelS2
One Speen Street
Framingham MA 01701 USA
www.LenelS2.com
Technical Support: +1 508.663.2505
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Introduction

The information in this hardening guide applies to NetBox™ Release 5.4.1 and later. The types of information covered in the guide are described below.

For the most recent version of the NetBox Hardening Guide, please see the version posted to the LenelS2 Portal at <https://portal.lenels2.com>.

Note:

Although VRx™ hardening guidelines are not addressed specifically in this edition of the Hardening Guide, many of the guidelines provided for NetBox also apply to VRx.

Chapter 1: Hardening Overview

Chapter 1 covers hardening basics and prerequisite skills and identifies industry-accepted tools and guidelines.

Chapter 2: Hardening Fundamentals

Chapter 2 provides hardening guidelines for deploying a NetBox access control system—including the network, client systems, and communication protocols that should be addressed prior to installing and hardening the system.

Chapter 3: Application Hardening

Chapter 3 provides hardening guidelines for specific settings that increase the security of a NetBox system.

Appendix A: System Diagram

Appendix A provides a diagram depicting the installation of a NetBox system in a hardened environment.

Appendix B: Port Usage

Appendix B identifies and defines the network ports used by NetBox systems.

Appendix C: Security Checklist

Appendix C describes the Security Checklist configuration report, which allows users with full system setup privileges to view the status of properties that affect NetBox system security.

Appendix D: LenelS2 Node Hardening

Appendix D provides instructions for hardening your LenelS2 nodes by requiring that connections to each node are secured with SSL, and by enabling secure communications with the Network Controller for each node.

Appendix E: Acronyms Used in this Guide

Appendix E provides a list of the acronyms used in this guide.

Chapter 1: Hardening Overview

This guide has been created primarily for NetBox system administrators and IT and security administrators who are responsible for the technical aspects of securing network appliances. It is to be used as a companion document to the *NetBox Hardware Installation Guide* (#NB-HW-*nn*, where *nn* refers to the latest revision).

The guide covers the steps and guidelines that should be implemented and maintained in the NetBox environment, with the goal of placing it in a recommended secure state. The guidelines were developed using highly recognized industry standards, applicable third-party recommendations, and security hardening best practices.

Note:

Each installer/system administrator should understand the system usage requirements and compliance directives to determine the hardening configuration applicable to client requirements. Take your specific environment into consideration before applying recommendations, and make efforts to secure additional elements of your environment as well.

The practices recommended in this document are designed to help mitigate the risks associated with network appliances. They build on and assume the implementation of practices described in the NIST publications on system and network security.

All customer IT regulations should be considered when applying the hardening guidelines covered in this document. Some guidelines will require that you ensure:

- All patching is current.
- Virus detection, malware and personal firewall software is installed on all desktop, laptop, and other client computers.
- Strong password and least-privilege policies are enabled.
- Sufficient logging is enabled.
- Any system default that does not yet conform to security best practices is changed to a more secure setting.

Before proceeding with the hardening of any system, be aware of the following key points:

- Understand what you have and how it is at risk. It is recommended that you perform an information risk assessment (in-house or via an independent expert) that looks at both technical and operational issues related to the security of your environment.
- Hardening standards should not be construed as one-size-fits-all. Every network and server is different. It all depends on your line of business, the regulations you are governed by, the risks, and the criticality of the NetBox appliance and the information it stores and/or processes.
- The customer's NetBox environment may contain necessary third-party or customer-specific integrations that could be affected when the system is hardened.

Prerequisite Skills

The material in this document is technically oriented. It is assumed that readers have at least a basic understanding of the following:

- System and network security.
- Installation of software on server and client computers.
- Operating systems, including configuration and management tools.
- Protocol enabling and disabling, ciphers, hashes, and key exchange algorithms.

Recommended Industry Tools

Throughout this document, there are references to common third-party or open source tools that might be useful during a particular hardening step.

These tools should be considered examples only. They are neither provided by nor supported by LenelS2. Review any such tool carefully and make choices based on your company policies. Any issues resulting from the use of such tools should be directed to the third-party tool vendors.

Steps to Hardening the NetBox Environment

Hardening your system configuration is only one step in securing your NetBox environment. Following the hardening steps below should be part of a comprehensive “defense-in-depth” security plan:

1. Ensure that all servers are installed in a physically secure environment (all servers must be installed in secure racks within a secure facility, and must include access monitoring and access logging into the secure racks and facilities).
2. Use strong passwords.
3. Follow the principle of least privilege (PoLP) when granting privileges and segmenting duties for user accounts.
4. Ensure that all critical servers and services are part of your business continuity and disaster recovery plan.
5. Employ appropriate network protection mechanisms (for example, firewall, packet filtering router, and proxy). The mechanisms you choose for a particular situation will depend on several factors. These include the location of the server's clients (for example, internet, internal, and remote access), the location of the server on the network, the types of services offered by the server, and the types of threats against the server.
6. Employ secure administration and maintenance processes. For example, ensure that the system data is backed up regularly and that software upgrades are applied.
7. Maintain proper alerts, records and logs. This information can be used, for example, to provide warnings of potential attacks, as legal and forensic evidence, and for the analysis of a recovery or continuous improvement plan.
8. Implement ongoing security training practices.
9. Do not:
 - Run any third-party integration or software on the NetBox system. The NetBox is an appliance, and only software from LenelS2 may be applied under the terms of the End User License Agreement.
 - Include unnecessary personally identifiable information (PII) in the NetBox database.

Secure by Default Features

To align with industry best practices for new installations, the NetBox features listed below are secure by default as of Release 5.4. System administrators should review these defaults on new systems to ensure they align with corporate policies.

Note:

Administrators should also review these defaults on the following systems, to make sure they align with recommendations in this guide and with corporate policies:

- NetBox systems upgraded to 5.4 or later from an earlier software version.
 - NetBox 5.4 systems whose system data was restored from an older system's backup file.
 - NetBox systems running software version 5.3.3 or earlier, before the strong defaults were adopted.
-

As of Release 5.4, the following features are secure by default:

- **SSL configuration:** When the network configuration is changed from the default for a new NetBox, VRx, or NetBox Global system, HTTPS is automatically enabled on the Network Settings page and a self-signed SSL certificate is generated.

Note:

Note: HTTPS is not enabled automatically for IP cameras. However, it is recommended that customers enable HTTPS for their cameras.

- **Transport Layer Security (TLS):** TLS v1.3 is now enabled by default for HTTPS on most platforms, including currently shipping products. All platforms support TLS v1.2, including older platforms.
- **Passwords:** Certain password policies and settings are now enabled by default. For more information, refer to [Maintaining Accounts and Passwords](#) on page 15.
- **API Logins:** As a security enhancement for session management, the web-based NetBox API (NBAPI) now returns a hash value rather than an integer when a user logs in. This value must be carried through to all subsequent requests.

Note that some users (such as those working in a language that does not permit an alphanumeric hash to be stored in an integer) might need to allocate a different and longer value to store the session ID.

In addition, the following security updates apply to Release 5.4 and later NetBox systems:

- **Login screen:** A notice appears on the screen indicating that proceeding with the login will create session cookies, which will be sent to and stored on the user's computer. If the user has reached the NetBox system using HTTP, a security warning recommending the use of HTTPS also appears on the screen.
- **Network Settings page:** Allows system administrators to configure all network settings on a single page. For example, they can change the Network Controller's IP address, enter its FQDN/CN, change its SSL Requirement setting, and upload a CA-signed SSL certificate rather than use the provided self-signed certificate.
- **Security Checklist Report:** This report allows users with full system setup privileges to view the status of various properties that affect system security.

Any property whose status does not conform to security best practices is flagged, and the report offers recommendations for addressing the issue. For more information, see [Appendix C: Security Checklist](#) on page 39.

Industry Accepted Guidelines

The following organizations publish common industry-accepted standards that include specific guidelines relevant to system hardening:

- [Center of Internet Security \(CIS\)](#)
- [International Organization for Standardization \(ISO\)](#)
- [National Institute of Standards and Technology \(NIST\)](#)
- [Open Web Application Security Project \(OWASP\)](#)
- [SysAdmin, Audit, Network and Security \(SANS\) Institute](#)

Architectural Assumptions of Scope

The architectural guidelines that follow identify the environmental conditions used during the development and testing of this guide. Because NetBox can be deployed in many and varied environments, you may find that your environment has elements not referenced within this scope.

Take your specific environment into consideration before applying recommendations and make efforts to secure additional elements of your environment as well.

Updates to this document may be published periodically as the NetBox product evolves and as new software releases are commercialized for use.

The scope of this guide includes:

- The NetBox environment with services, servers, and access control devices based on support included with NetBox 5.4. Video support and devices are not included at this time.

To support integration with the customer's systems and third-party services, NetBox provides an API (NBAPI) for external application integration and a Data Operations tool for scripted import and export operations. When using these integration methods, it is the responsibility of the entity making the connection to ensure that security best practices are observed to protect the data entered into or removed from the system.

Hardware Scope

When setting up LenelS2 nodes, we recommend requiring that all connections to the node are secured with SSL. We also recommend enabling secure node communications to allow nodes to exchange data securely with the Network Controller. For instructions, see [Appendix D: LenelS2 Node Hardening](#) on page 42.

For any third-party device, refer to the vendor documentation for information on hardening the device.

Software Scope

All NetBox security functions are delivered through a built-in web user interface. NetBox employs fully integrated operation with the following thick client applications to support live video streaming and forensics:

- Magic Monitor[®]
- Magic Video Player (MVP)

These applications can be individually licensed to meet the needs of the user.

To reduce the attack surface of your NetBox system, install only the thick client applications required to support the user at a given location.

For more information on the NetBox environment, refer to [Appendix A: System Diagram](#) on page 23 and [Appendix B: Port Usage](#) on page 24.

Note that the NetBox Global web user interface was implemented using Microsoft Silverlight, which will reach the end of support in October 2021. The interface is currently being replaced by a web user interface for all configuration and cardholder administration functions. Magic Monitor 7.6 can be installed to deliver Global Security Operations Center (GSOC) security monitoring functions. Magic Monitor 7.7 will add cardholder administration to complete the GSOC functionality.

Chapter 2: Hardening Fundamentals

This chapter provides hardening guidelines for areas outside the NetBox system, such as the network, communication protocols, and client systems that should be addressed prior to installing and hardening the NetBox system.

Security Checklist

In Release 5.4, a new configuration report, Security Checklist, was added to the NetBox application. Users with full system setup privileges can use the report to view the status of properties that affect system security.

For more information, refer to [Appendix C: Security Checklist](#) on page 39.

Protocols

Network protocols are sets of established rules that dictate how to format, transmit and receive data so that computer network devices—from servers and routers to endpoints—can communicate regardless of the differences in their underlying infrastructures, designs, or standards. Network security protocols, such as TLS/SSL, are primarily designed to prevent any unauthorized user, application, service, or device from accessing network data. This protection applies to virtually all data types, regardless of the network medium used.

Digital Certificates

To best support TLS/SSL, a strong private key is needed to prevent attackers from carrying out impersonation attacks. It is equally important to have a valid and strong digital certificate, which will give the private key the right to represent a particular hostname.

Administrators should generate unique TLS/SSL keys for each device and store these keys in secure ways to prevent unauthorized access. Consider using a hardware security module (HSM) to secure digital keys.

To manage TLS certificates in the NetBox system, select Configuration  on the page bar, navigate to *Site Settings: Network Controller*, and select **Update SSL Certificate** on the Web Site tab. The best practice is to select the *SSL Upload Certificate from CSR* option. With this option, the private key is generated and remains on the NetBox.

Note that the *SSL Upload Certificate and Key* option should be selected only if it is absolutely required by your IT policies. It is your responsibility to follow the policy your IT organization has set up to protect the key file.

Printer Security

NetBox allows the web browser to print UI pages and report output on networked printers. When implementing security best practices, printers with network access should be treated the same way as any other endpoint.

Network printers that are not hardened can be targets of cyber attacks. This is because such printers:

- Often fall out of focus for IT departments.
- Are more sophisticated and have many more capabilities than their predecessors, such as the abilities to scan and fax.
- Possess operating systems and control capabilities similar to computers.
- Often go without hardening and patching that would be required of other such capable endpoints.
- Are shipped and installed with many open default ports, such as common TCP and UDP ports.

If networked printers are not hardened, attackers can gain access through wireless access points, or from infected computers. After gaining access, an attacker can scan for exploitable systems and pivot to other connected machines on the internal network, and/or cause a Denial of Service (DoS) attack. An attacker can also gain access to the printer's print spool, which may contain sensitive information.

Hardening Best Practices for Networked Printers

- Inventory all printer endpoints and update them with the latest patches available.
- Change all default passwords and ensure that all passwords are encrypted.
- Disable any rarely used or extraneous services, such as File Transfer Protocol (FTP).
- Select management protocols that provide encryption (HTTPS or SSH).
- If possible, segment networked printers from enterprise systems by placing the printers on a VLAN, thus preventing them from having access to the entire internal network. For more information see [Isolating the NetBox Resources Within a VLAN](#) on page 22.
- If possible, place networked printers behind a firewall to limit access.

Client-side Protections

Note that some of the recommendations below are specific to Microsoft Windows. However, similar guidelines exist for other operating systems, including OSX and most Linux desktops.

To protect against malicious attacks or vulnerabilities, it is important to ensure that the following best practices for client-side protections are in place:

- All clients should be protected and within your network and should be actively monitored.
- Restrict physical access to workstations to authorized personnel only. It is recommended that all workstations remain attended. If unattended workstations are required, include only those that are absolutely necessary.
- Install the Magic Monitor and Magic Video Player clients only on specific workstations, thereby limiting the install base to the fewest number of workstations needed.

- Run the Magic Monitor and Magic Video Player clients only on physically secure workstations. Only the display, and if necessary the keyboard, should be in an exposed location. If these clients must be run on workstations in exposed physical spaces, such as in a lobby or unmanned monitoring kiosk, consider the use of virtual terminal services to further protect such endpoints.
- The first line of defense is denying local administrator access to workstation users, preventing attackers from taking advantage of user privileges in order to install malware or attack tools.
- Restrict users' ability to temporarily disable anti-virus protection.
- Ensure that users are aware of the dangers of clicking on unknown links and consenting to the installation of unauthorized software.
- Enact strong complex passwords, following the recommendations in [Guidelines for Creating Strong Passwords](#) on page 16.
- Conduct an examination of your anti-virus solutions and ensure that:
 - Antivirus monitoring is enabled on all workstations.
 - Rules are enacted to prevent installation of potentially unauthorized software.
 - Your antivirus solutions include protections against process memory injections and provide Memory Exploit Mitigation (MEM).

Note that most freeware antivirus software will not remove the tools that can be used for further memory exploits. This is important, because kernel malware circumvents this abstraction of privileges by running in kernel mode with direct access to all system services. In other words, it has complete control of the infected system. One attack vector is the installation of a malicious driver.
 - Your antivirus software does not whitelist kernel mode process tools such as Process Hacker.
 - Your antivirus solutions leverage heuristics for further protection.
- Disable all USB access to prevent loading of unapproved software that can be used to perform attacks. It is recommended that USB ports are disabled both through policy settings and through physical port blockers.
- Implement firewalls on all workstations. Configure each firewall to block all incoming connections, from the internet and the local area network, to services that should not be publicly available. By default, you should deny all incoming connections and allow only services you explicitly want to offer to the outside world. This can help prevent self-propagating infections from spreading.
- Implement a whitelist for all outbound connections.
- Implement a strong patch management process. Patching helps eliminate software flaws that can be used to inject malicious kernel code.

- On Microsoft Windows, disable AutoPlay to prevent the automatic launching of executable files on the network and on removable drives, and disconnect removable drives when they are not required. If write access is not required, enable read-only mode if the option is available.
- Unless file sharing is required, turn it off. If file sharing is required, use ACLs and password protection to limit access. Disable anonymous access to shared folders. For folders that must be shared, grant access only to user accounts with strong passwords.
- Enable SMB signing.
- Disable SMBv1.
- If Web Proxy Auto-detect is not in use, disable it.
- Configure your email server to block or remove email containing file attachments that are commonly used to spread threats, such as .vbs, .bat, .exe, .pif and .scr files.
- Train employees not to open attachments unless they are expecting them. In addition, do not execute software that is downloaded from the internet unless it has been scanned for viruses. Simply visiting a compromised website can cause infection if certain browser vulnerabilities are not patched.

Chapter 3: Application Hardening

This chapter provides guidelines for hardening the NetBox application. It includes information on:

- Maintaining accounts and passwords.
- Enabling secure web browser communications ([page 18](#)).
- Completing the Network Controller setup from a single location ([page 19](#)).
- Enabling ODBC encryption ([page 20](#)).
- Refreshing activity logins ([page 21](#)).
- Backup best practices ([page 21](#)).
- Isolating the NetBox Resources within a VLAN ([page 22](#)).

To see how NetBox is installed in a hardened environment, refer to [Appendix A: System Diagram](#) on page 23.

Maintaining Accounts and Passwords

Administrator Passwords

Immediately after installing a NetBox appliance or LenelS2 node, you should change its default administrator login password. For instructions, refer to the *NetBox Initial Software Setup Guide* (#NB-SS-*nn*, where *nn* refers to the latest revision).

Thereafter, the administrator password should be changed at least every 30-60 days.

User Passwords

"After giving users login rights to a NetBox system, which includes assigning them passwords, the user can be forced to change the password on first login. To enable this, select Configuration  on the page bar and navigate to *Site Settings : Network Controller*. On the Admin tab, check the *Force password change on first login* box.

It is recommended that you make sure each user changes his or her password at least every 90 days.

By setting an expiration period for passwords, you can ensure that each user will be required to change his or her password after the specified number of months have elapsed. To do this, select Configuration  on the page bar and navigate to *Site Settings : Network Controller*. On the Web Site tab, specify the number of months on the *Password expiration* drop-down list. You can also specify a minimum password length, and you can require that passwords contain letters, numbers and special characters.

Guidelines for Creating Strong Passwords

Use the following guidelines when creating passwords:

- Always consult your corporate password policies, which may require that passwords contain letters, numbers, and special characters.
- When changing a password, do not reuse a password used previously on that system or on any other system. Each password should be unique across all systems.
- A password should not contain the login username or parts of the user's full name, such as the first or last name.
- A strong password should always be at least 8 characters long, and ideally it should be longer than 14 characters.
- Consider requiring that passwords for administrators or users with privileged roles be at least 14 characters long.
- Using passphrases rather than passwords is recommended.

A passphrase is a collection of common words combined randomly into a sentence or phrase. Because it is longer than a password and includes multiple words, a passphrase provides greater protection against brute force and dictionary attacks. Randomly combining bizarre and uncommon words into a sentence or phrase creates an even stronger password.

Note:

To check your system's password health at any time, view the Security Checklist configuration report. Refer to [Appendix C: Security Checklist](#) on page 39.

Setting Up Active Directory Configurations for Authenticating System Users

If your organization uses an LDAP server for password authentication, user login credentials are stored in a centralized, network-based repository that is accessed using LDAP. By configuring LDAP user authentication in the NetBox, you can enable users to log in using their LDAP directory service credentials, which are authenticated by the directory service.

Once LDAP user authentication is configured, NetBox administrators will configure each user's person record to use the LDAP directory service rather than enter NetBox login credentials for the user.

Note:

The directory services supported for user authentication are Microsoft Active Directory (AD) and OpenLDAP.

To configure LDAP user authentication in the NetBox, select Configuration  on the page bar, navigate to *Network Resources : Active Directory Integration*, and create a configuration for authenticating users.

By default, LDAP traffic is transmitted unsecured. When creating the user authentication configuration, it is recommended that you protect connections to the directory service over LDAP by specifying that the LDAP server will use a security protocol (TLS/SSL or LDAPS) for these connections. When the server uses one of these protocols, data is encrypted when it is sent across the network and will not be vulnerable if it is intercepted.

The available options are:

- **Standard Transport Layer Security.** This is the preferred protocol. The server uses secure LDAP connections through a configurable port (port 389 by default). When a user presents his or her login credentials, the server first establishes an unencrypted connection with the client machine, then starts the TLS layer using the StartTLS operation.
- **LDAP over SSL.** The server uses secured LDAP connections through a configurable port (port 636 by default). When a user presents his or her login credentials, the server establishes an SSL-encrypted connection with the client machine.

Note:

If the LDAP connection port is configured to a port other than one of the default ports (389 or 636), the AD server must also be configured to use the same port. Before configuring a different port, consult your network administrator to ensure you can maintain a connection to the AD server.

Displaying a Privacy Statement

To display a Privacy statement on the login page for NetBox, navigate to **Configuration : Site Settings : Network Controller** and select the **Web Site** tab, and then enter the privacy statement in the **Privacy** text box. You can enter a combination of text and HTML. For example, you can use HTML to create a bold title or add a line break. The character limit is 1024 characters.

Note:

If your Banner or Privacy statement contains line breaks, the entire text may not be visible in a normal-sized browser window. A scrollbar will appear on the page to notify users that more text appears below the visible text in the current view.

Enabling Secure Web Browser Communications

It is recommended that you enable secure communications between the Network Controller and the web browser by doing the following:

- **ConfigureSSL**. Configuring the system's embedded web server to use an SSL certificate ensures that data sent between the Network Controller and the web browser is protected, either by encryption alone or by encryption plus authentication.

It is recommended that you use the Network Settings page to change the SSL Requirement setting to either redirect HTTP to HTTPS or allow only HTTPS. This will prevent users from inadvertently using unencrypted browser communication to the NetBox.

- **Enable HTTPS**. Enabling the HTTPS communication protocol ensures that data exchanged between the Network Controller and the web browser is transmitted over a secure channel, providing protection from eavesdroppers and man-in-the-middle attacks.

If HTTPS was not enabled prior to the upgrade, it is recommended that you enable it by generating a self-signed SSL certificate manually or by uploading an SSL certificate.

Beginning with Release 5.4, secure web browser communications are enabled by default on new NetBox, NetVR, and NetBox Global systems, and on upgraded systems for which they were enabled previously.

- For a new system, changing the factory default IP address automatically enables HTTPS and generates a self-signed SSL certificate that is signed with your web server's own private key. You can use the Network Settings page to change the system's digital certificate and SSL Requirement setting, as described in the following section.
- For an upgraded system on which HTTPS was enabled previously, HTTPS remains enabled, and the same SSL certificate is used.

On new systems, HTTPS is not enabled automatically for IP cameras. However, it is recommended that customers enable HTTPS for their cameras.

If you disable HTTPS, the existing SSL certificate is not deleted, and it will be used if HTTPS is re-enabled later.

If the existing SSL certificate is compromised, you can generate a new self-signed certificate on the Network Settings page, as described in the following section.

Completing the Network Controller Network Setup

To complete the Network Controller network setup from a single location, select Configuration  on the page bar, navigate to *Site Settings : Network Settings*, and use options on the Network Settings page to:

- Enter the Network Controller's Fully Qualified Domain name (FQDN), also known as its Common Name (CN). This will allow users to log in using the FQDN/CN rather than its IP address.
- Enter the HTTPS port used for secure web browser access to the system.
- Update the system's digital certificate. Available options for configuring SSL certificates are:
 - **SSL Self-Signed**. This simple and cost-free option lets you generate a certificate that is signed with your web server's own private key.
 - **SSL Upload Certificate from CSR**. This option lets you generate a certificate signing request (CSR), which you can send to a certificate authority (CA), and then upload the trusted certificate provided by the CA.
 - **SSL Upload Certificate and Key**. This option lets you upload your organization's own certificate and matching key. Optionally, you can also upload an intermediate CA file (sometimes called a chained certificate) that links your certificate to a trusted root certificate.

For more information on configuring SSL certificates, refer to the NetBox online help. (Search for *Enabling Secure Web Browser Communications*.)

Note:

SSL certificates and keys are not included in system backups. Following the restoration of a system backup or a reset of the Network Controller to factory defaults, you will need to generate or upload the certificates and keys that were generated or uploaded previously.

- Change the system's SSL Requirement setting. Available options are:
 - **HTTPS only** (disables http). HTTP is disabled, and HTTPS is used for all web browser communications.
 - **Redirect HTTP to HTTPS**: All http:// requests are redirected to https://.
 - **Allow HTTP or HTTPS**. Both HTTP and HTTPS are enabled and can be used for web browser communications. Because this setting makes it easy for users to connect via unencrypted HTTP, is not recommended.
 - **HTTP only (insecure)**. HTTPS is disabled and only HTTP is used for web browser communications. Because HTTP is unencrypted, this setting is not recommended.

The HTTP only option is enabled on systems that have been upgraded to Release 5.4 or later and previously used HTTP for web browser communications. As a security best practice, it is recommended that administrators select a more secure option.

Important:

In addition to enabling secure communications with web browsers, we recommend enabling secure communications with LenelS2 nodes. For instructions, refer to [Appendix D: LenelS2 Node Hardening](#) on page 42.

Enabling ODBC Encryption

By enabling Open Database Connectivity (ODBC), you can allow users to connect directly to the Network Controller using ODBC and create reports from the NetBox database. While connected, users are permitted to read only information that is relevant to reporting.

To enable ODBC, select Configuration  on the page bar and navigate to *Site Settings: Network Controller*. On the Data Integration tab, select the Enabled check box under ODBC.

When enabling ODBC, you should select the SSL check box to enable ODBC encryption, and enter the password a user will need to enter to use this feature. Rather than use the default password shown on the page, enter a new, complex password, following the recommendations in [Guidelines for Creating Strong Passwords](#) on page 16.

ODBC is currently supported through a self-signed certificate that is unique to each NetBox. For that reason, when connecting to the Network Controller from an ODBC client application, users might need to ignore Certificate Validity and Constraints warnings. They will also need to allow the client to accept the self-signed certificate generated by the NetBox system.

Refreshing Activity Logins

Keeping browser sessions alive indefinitely increases your organization's exposure to session-based attacks by giving attackers sufficient time to exploit existing security vulnerabilities. For example, anyone with access to a valid session identifier could record it offline and use it later to log into the NetBox.

As a security best practice, you may want to consider configuring a shorter web browser session duration for user logins. Consult your IT administrator if you need assistance.

Backup Best Practices

It is essential to have a backup strategy in place for your NetBox systems that is in line with the latest technology and best practices. Your strategy should define, for example, what data is retained; how, and with what frequency, the data is backed up; and how backups are stored.

For information on backing up your NetBox system, refer to *Backing Up the System Data* in the NetBox online help.

Note:

To check your system's backup health at any time, view the Security Checklist configuration report. Refer to [Appendix C: Security Checklist](#) on page 39.

Isolating the NetBox Resources Within a VLAN

It is common for organizations to run all of their network infrastructure on a topology that handles security segmentation through a few well-placed firewalls (such as public-to-DMZ-to-Internal-to-Higher Security).

Another approach offering significantly higher security is the utilization of a Virtual Local Area Network (VLAN). By leveraging the capabilities of managed switches, a VLAN allows a subset of the network infrastructure to operate on a network that is normally isolated from all other networks.

There are several security benefits when using a VLAN as another layer of defense to isolate your NetBox resources. For example, using a VLAN:

- Significantly reduces the attack surface against attackers who have penetrated parts of the local network.
- Restricts access to sensitive data by placing only those users who have access to the data on the VLAN, thus reducing the chances of an outsider gaining access to the data.
- Controls broadcast domains, sets up firewalls, restricts access, and informs the network manager of an intrusion.
- Helps defend the NetBox application against threat vectors inherited from older Microsoft® and third-party technologies that do not have updated replacements.

Because a VLAN is highly dependent on the vendor, the selection of network equipment, and local topologies, its design and deployment is not covered in this document. However, in general terms, the standard practice may consist of the following steps:

1. Define a new (private or isolated) VLAN using your network administration tools.
2. If the VLAN is not isolated, configure a DMZ rule for VLAN access; for partial isolation, configure a jump server.
3. Migrate all the clients, servers and network-attached devices to the VLAN.
4. Provide a separate VLAN, or physical stub networks, for cameras so they cannot communicate directly with the internet.

Appendix A: System Diagram

The following system diagram depicts a NetBox configured in an environment with multiple components attached or configured. These include NetBox Global, VRx, and NetVR, as well as several independent lock types and panels.

Because each environment is configured according to the customer's needs and implementation strategy, you should focus on the components within your own specific environment when viewing the diagram.

For additional, optional port configurations, refer to [Appendix B: Port Usage](#) on page 24.

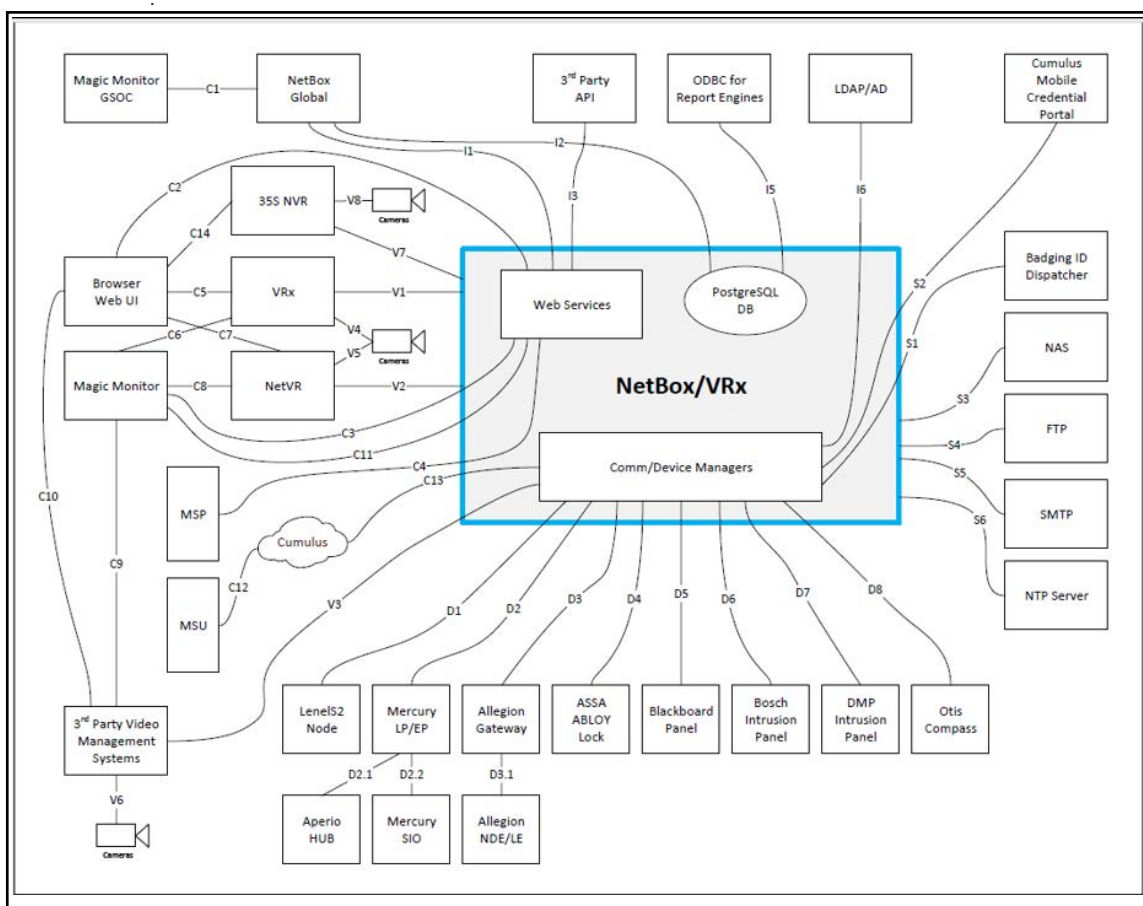


Figure 1. NetBox system in a hardened environment.

Appendix B: Port Usage

By securing the network ports used by your NetBox system and closing unused ports, you can mitigate possible security risks associated with these ports.

Note:

Ports used by NetBox are open only if the associated devices or features are licensed. Otherwise, these ports are hidden via UFW-configured iptables firewall rules, which make them unresponsive to port scans.

Tables in the following sections list the ports used for:

- Client connections ([page 25](#))
- Video integration connections ([page 27](#))
- Mobile Security User (MSU)/Cumulus Mobile App WAN connections ([page 29](#))
- NetBox Global operations integration ([page 30](#))
- Customer data integrations ([page 31](#))
- Badge printing services ([page 33](#))
- IT services ([page 34](#))
- Device management communication ([page 35](#))

For each port listed in a table, the number in the leftmost column refers to the corresponding network connection depicted in [Appendix A: System Diagram](#) on page 23.

Client Connections

Table 1. Ports Used for Client Connections.

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
C1, C2, C3, C4	• Browser/Magic Monitor • Web Services/REST	• Browser • Mobile Security Professional (MSP) - NetBox only • Magic Monitor	• NetBox • Global	Assa locks call-in on Port 2571, which must be open to the Network Controller for communications with Assa locks. Firewall path from NetBox to Assa PoE locks running in legacy mode(LenelS2 NetBox Mode in LCT tool) must be open. Ports are configurable by users with the Full system setup role.	Configurable: • 443 - HTTPS • 80 - HTTP Fixed: • 2571 - TCP • 8023 -TCP	HTTPS - TLS v1.2, TLS v1.3 Note that: • HTTP port can be disabled. • HTTP can be redirected to HTTPS. • HTTPS/HTTP ports can be configured.
C5, C6	VRx video, live and recorded	• Magic Monitor • Browser (video) • MSP (video) - not illustrated	VRx	Must be open from client directly to the VRx that manages selected camera and stores related video HTTPS/HTTP ports are configurable by users with the Full system setup role.	Configurable: • 443 - HTTPS • 80 - HTTP Fixed: 554 - RTP	• HTTPS - TLS v1.2, TLS v1.3 • RTP - none Note that: • HTTP port can be disabled. • HTTP can be redirected to HTTPS. • HTTPS/HTTP ports can be configured.

Table 1. Ports Used for Client Connections. (Continued)

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
C7, C8	NetVR video, live and recorded	• Magic Monitor • Browser (video) • MSP (video)	NetVR	Must be open from client directly to the NetVR that manages selected camera and stores related video HTTPS/HTTP ports are configurable by users with the Full system setup role.	Configurable: • 443 - HTTPS • 80 - HTTP Fixed: • 22609 - TCP • 4503 -TCP (MSP only)	None/simple authentication
C9, C10	Third-party video, live and recorded	• Magic Monitor • Browser (video)	Third-party VMS	Must be open from client directly to the third-party VMS that manages selected camera and stores related video	Determined by VMS vendor's Implementation	None/simple authentication
C11	NetBox - Magic Monitor command callback	NetBox	Magic Monitor	Must be open from the NetBox/VRx to allow it to trigger Magic Monitor view changes	1080 - Proprietary	None/simple authentication
C14	35 Series NVR	Browser (video)	35 Series NVR	Must be open from client directly to the 35S NVR that manages selected camera and stores related video HTTPS/HTTP ports are configurable by users with Administrator role.	Configurable: • 443 - HTTPS • 80 - HTTP	HTTPS - TLS v1.2

Video Integration Connections

Table 2. Ports Used for Video Integration Connections.

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
V1	NetBox - VRx	Bidirectional	Bidirectional	VRx VMS integration commands from the NetBox to the VRx use these ports via proprietary protocol Events from the VRx to the NetBox use these ports via proprietary protocol	• 41000, 41001, 41002 - TCP • 123 - UDP from VRx to NetBox NTP server	None/simple authentication
V2	NetBox - NetVR	Bidirectional	Bidirectional	NetVR VMS Integration NetBox->Port 3011->NetVR NetVR->Port 3010->NetBox Commands from the NetBox to the NetVR use port 3011 via a proprietary protocol. Events from the NetVR to the NetBox use port 3010 via a proprietary protocol.	• 3011 - TCP • 3010 - TCP • 123 - UDP from NetVR to NetBox NTP server	None/simple authentication
V3	NetBox - third-party video	Bidirectional	Bidirectional	Third-party video support NetBox->Port 3011->VMS VMS->Port 3010->NetBox Commands from the NetBox to the VMS use port 3010 via proprietary protocol Events from the VMS to the NetBox use port 3010 via proprietary protocol	• 3011 - TCP • 3010 - TCP • 123 - UDP from third-party VMS to NetBox NTP server	None/simple authentication

Table 2. Ports Used for Video Integration Connections. (Continued)

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
Note: Ports not specified as configurable are fixed.						
V4, V5, V6, V8	Camera - VRx/NetVR/35S NVR or third-party VMS	VRx/NetVR /35S NVR opens connection for video streaming and camera control	Camera accepts connection for video streaming and camera control	HTTPS/HTTP must be open to camera for camera control by VRx/NetVR/35S NVR Real Time Streaming uses RTSP. Port 554 must be open to camera to receive video streams on the VRx/NetVR Bidirectional NTP Port 123 - UDP must be open to synchronize camera time Bidirectional Zeroconf/WS-Discovery may be open to LAN/VLAN segment to simplify initial cameras setup	• 443 - HTTPS • 80 - HTTP • 554 - RTP/RTSP • 123 - UDP for NTP • 5353 - UDP Zeroconf • 3702 - UDP WS-Discovery	• HTTPS supports TLS v1.2, TLS v1.3 • RTP/ RTSP video streaming - None • NTS not supported • Zeroconf/WS-Discovery - None
V7	NetBox - 35S NVR	Bidirectional	Bidirectional	35S VMS integration commands, data queries, and events to and from the 35S NVR	• 80 - HTTP • 443 - HTTPS • 30143 - TCP	HTTPS - TLS v1.2

Mobile Security User (MSU)/Cumulus Mobile App WAN Connections

Table 3. Ports Used for Mobile Security User (MSU)/Cumulus Mobile App WAN Connections.

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
C12	Cumulus - MSU	MSU	Cumulus	MSU uses Cumulus as relay to exchange info with NetBox.	443 - HTTPS	TLS v1.2, TLS v1.3
C13	NetBox (Comm/Dev Managers) - Cumulus	NetBox	Cumulus	NetBox uses Cumulus as relay to exchange info with MSU	443 - HTTPS	TLS v1.2, TLS v1.3

NetBox Global Operations Integration

Table 4. Ports Used for NetBox Global Operations Integration.

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
I1, I2	NetBox Global cardholder synchronization and event/activity aggregation	Bidirectional	Bidirectional	Must be open to the Network Controller to receive communication from devices. WirelinePoE locks also support connections initiated from the NetBox. Port 9090 must be open on the NetBox Global server. It is used for NetBox to NetBox Global communications.	Configurable: • 443 - HTTPS • 80 - HTTP	HTTPS - TLS v1.2, TLS v1.3 Note that: • HTTP can be disabled. • HTTP can be redirected to HTTPS. • HTTPS/HTTP ports can be configured.
(not illustrated)	Network Storage for backups				SMB, SSH	
(not illustrated)	NTP Server	Global	NTP		123 - UDP	NTS not supported

Customer Data Integrations

Table 5. Ports Used for Customer Data Integrations.

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
I3	• NBAPI real-time integration • Data Operations batch integration	Custom HTTP Client	• NetBox • Global	Must be open to the Network Controller. Ports are configurable by users with the Full system setup role.	Configurable: • 443 - HTTPS • 80 - HTTP	HTTPS - TLS v1.2, TLS v1.3 Note that: • HTTP port can be disabled. • HTTP can be redirected to HTTPS. • HTTPS/HTTP ports can be configured.

Table 5. Ports Used for Customer Data Integrations.

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
I5	NetBox third-party report engine ODBC integration - ODBC PostgreSQL	Custom report engine	NetBox	For PostgreSQL report usage	5432 - ODBC (TCP)	SSL option
I6	LDAP/AD authentication and cardholder data management	NetBox	Company LDAP/AD server	Must be open to the Network Controller with appropriate account access granted by server	Configurable: • 389 - TCP/UDP LDAP • 636 - TCP LDAP SSL Note: If the LDAP connection is configured to a port other than 389 or 636, the AD server must also be configured to use the same port. Before configuring a different port, consult your network administrator to ensure you can maintain a connection to the AD server.	SSL option

Badge Printing Services

Table 6. Ports Used for Badge Printing Services.

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
S1	NetBox (Comm/Dev Managers) - Badging ID Dispatcher	NetBox	Print and capture stations	Must be open to Network Controller for communications between the Badge Server and the NetBox TCP ports are configurable by users with the Full system setup role.	Configurable: • 81- TCP • 8080- TCP (notifications)	• HTTPS option available • TLS v1.3

IT Services

Table 7. Ports Used for IT Services.

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
S2	NetBox - Cumulus Mobile Credentials	NetBox	Cumulus	Heath monitoring, mobile credential management	443 - HTTPS	TLS v1.2, TLS v1.3
S3	NetBox - SMB Network Storage Global - SMB Network Storage	NetBox, VRx, Global	NAS Server	Capture backup files, archive files, and recorded video	• 445 - SMB • 22 - SSH (NetBox Global only) (139 legacy NetBIOS will be phased out)	Standard SMB v2 security options
S4	NetBox - FTP Network Storage	NetBox	FTP Server	When active FTP is used, these ports must be open to FTP server for FTP backups from the Network Controller. When passive FTP is used, port 20 will not be required. Ports must also be left open to the Network Controller for FTP server responses. The network administrator must set up these ports.	20, 21 - TCP	SFTP supported with simple authentication
S5	NetBox - SMTP Server	NetBox	SMTP Mail Relay Server	SMTP access email relay (for the event action "Send Email")	• 25 - TCP • 587 - TCP	Port 587 offers encryption options with simple authentication
S6	NetBox - NTP Server	NetBox, VRx, Camera	NTP Server, NetBox, VRx	NTP (Network Time Protocol) - used for time synchronization	123 - UDP	NTS not supported

Device Management Communication

Table 8. Ports Used for Device Management Communication.

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
D1	NetBox - LenelS2 Network Node	Network Node	NetBox	Network Node call-in on Port 7262, which must be open to the Network Controller for communications with LenelS2 Nodes NetBox HTTPS Port must open from node for firmware updates. HTTP will be used if HTTPS is not enabled. Be sure these ports are open through routers and firewalls for any LenelS2 nodes on different subnets than the Network Controller.	• 7262 - TCP • 443 - HTTPS (or 80 - HTTP)	TLS v1.2, TLS v1.3
D2	NetBox - Mercury	NetBox	Mercury panel	Firewall path from NetBox to each Mercury panel must be open	3001 - TCP	TLS (version may depend on Mercury firmware version)
D2.1	Mercury - Aperio HUB	Bidirectional (RS-485)	Bidirectional (RS-485)	Firewall path from NetBox to each Aperio Hub must be open	Proprietary RS-485	Private low voltage link
D2.2	Mercury - Mercury SIO	Bidirectional (RS-485)	Bidirectional (RS-485)	RS-485 communication between Mercury Panel and I/O cards	Proprietary RS-485	Private low voltage link
D3	NetBox - Allegion Gateway	NetBox	Allegion Gateway	Firewall path from NetBox to each Allegion Gateway must be open	80 - HTTP	Not supported

Table 8. Ports Used for Device Management Communication. (Continued)

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
D3.1	Allegion Gateway - NDE/LE locks	Bidirectional (900MHz RF)	Bidirectional (900MHz RF)	RF 900MHz wireless communications	Proprietary	Proprietary 900MHz communication
D4	NetBox - ASSA ABLOY locks	ASSA ABLOY lock	NetBox	Assa locks call-in on Port 2571, which must be open to the Network Controller for communications with Assa locks. Port 8023: Firewall path from NetBox to Assa PoE locks running in legacy mode(LenelS2 NetBox Mode in LCT tool) must be open.	2571 - TCP 8023 -TCP	AES option
D5	NetBox - Blackboard	Blackboard panel	NetBox	Blackboard device call-in Must be open to the Network Controller for communications with Blackboard nodes Be sure this port is open through routers and firewalls for any Blackboard nodes on different subnets than the Network Controller.	9005 - TCP	TLS v1.2
D6	NetBox - Bosch intrusion panels	NetBox	Bosch panel	Used for outbound communications between the NetBox system and supported BOSCH intrusion panels	7700 - TCP	TLS v1.2

Table 8. Ports Used for Device Management Communication. (Continued)

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
D7	NetBox - DMP intrusion panels	DMP panel	NetBox	DMP device call-in 2001 - (LenelS2 product outbound): Used for outbound communications between NetBox system and supported DMP intrusion panels running with firmware version 205 or earlier 2011 (LenelS2 product outbound): Used for outbound communications between NetBox system and supported DMP intrusion panels running with firmware version 206 or later 6000-6063 (inbound to LenelS2): Must be open to Network Controller for assignment to each DMP panel created in the NetBox system This port is needed as the PC Logging port for status communications to the NetBox system. It is the same for all DMP panels and cannot be changed.	• 2011 - TCP • 6000-6063 - TCP Note that older setups may have used 2001 - TCP rather than 2011 - TCP.	None

Table 8. Ports Used for Device Management Communication. (Continued)

Diagram Ref.	Description	From Client	To Server	Usage Notes	Port - Protocol	Encryption & Security
					Note: Ports not specified as configurable are fixed.	
D8	NetBox - Otis Compass	Bidirectional private link or VLAN	Bidirectional private link or VLAN	Required for Otis Elevator Compass integrations	• 45303, 45307, 46307 - UDP NetBox • 46307, 46303, 46308 - UDP Compass	None - Reference Design specifies private network link for communication

Appendix C: Security Checklist

Users with full system setup privileges can use the Security Checklist report to view the status of properties that affect system security. To open the report, select Administration  on the page bar, navigate to *Reports : Configuration*, and select Security Checklist on the Configuration Reports menu.

The report shows the status of the following properties:

- **Admin Account Password Age**. Indicates when the administrator login password was last updated.
- **Strong User Account Passwords**. Indicates the current system requirement for user account password length.
- **SSL Certificate (for https)**. Indicates whether an SSL certificate is configured for the system, and if so, indicates the certificate's expiration date.
- **ODBC Report Access**. Indicates whether ODBC is enabled for the system, and if so, indicates whether SSL encryption is enabled.
- **Backup Health**. Indicates when the last backup was saved.
- **Recent ssh activity**. Indicates the date of the last login via SSH and the username entered for each successful SSH login.

Asterisks displayed in the leftmost column flag properties whose status does not conform to security best practices. We include one or more asterisks to help the system manager prioritize issues that should be addressed.

- *** Three asterisks indicates an issue that poses a high risk of data disclosure or harm to the system and should be corrected promptly.
- ** Two asterisks indicates an issue that should be corrected to increase the effort required to attack the system. In some cases, the required changes might be inconvenient to implement (long passwords, for example). However, security best practices recommend the changes be implemented as soon as possible.
- * One asterisk indicates an issue that, if corrected, will improve the security of the system. In some cases, the level of risk might be acceptable to the end user, due to the effort required to implement the required changes, or because the risk is mitigated by the way the system is deployed.

Properties whose status does not conform to security best practices are flagged, and the report offers recommendations for addressing the issues

The example that follows shows how the status of flagged properties changes as the system administrator takes remedial actions.

After upgrading a NetBox system to software version 5.4, the system administrator views the Security Checklist report:

	Property 	Status	Recommendations
	Admin Account Password Age	Last updated 05/03/2021	
*	Strong User Account Passwords	Minimum password length is set to 8	Best practice for password length is 14 character or more.
***	SSL Certificate (for https)	Not configured	Configure SSL https with self-signed certificate or better
**	ODBC Report Access	Enabled without SSL encryption	Enable SSL encryption on NetBox and configure your ODBC clients to use it.
	Backup Health	Backup SAVED 2021-05-03T19:13:29+0000	
	Recent ssh activity	None	

Figure 1. Security Checklist report indicating three security issues.

Asterisks displayed in the leftmost column indicate the following security issues:

- The minimum password length for user accounts is set to 8 characters rather than the recommended 14 or more characters. This issue poses minimal risk of data disclosure, so it is flagged with a single asterisk.
- Because SSL was disabled prior to the upgrade, there is no SSL certificate configured. This issue poses a serious risk of data disclosure, so it is flagged with three asterisks.
- ODBC is enabled without SSL encryption. This issue poses a potential risk of data disclosure, so it is flagged with two asterisks.

Figure 2 shows the Security Checklist report after the system administrator has made the following changes:

- Set the minimum password length for user accounts to 14 characters. There are now no security issues associated with user account passwords.
- Generated a self-signed SSL certificate. However, because the administrator did not disable HTTP by changing the default SSL Requirement setting, the property is still flagged, this time with two asterisks.
- Enabled SSL encryption for ODBC. There are now no security issues associated with ODBC report access.

	Property 	Status	Recommendations
	Admin Account Password Age	Last updated 05/03/2021	
	Strong User Account Passwords	Minimum password length is set to 14	
**	SSL Certificate (for https)	Self-signed certificate expires Aug 6 20:23:18 2023 GMT http is not disabled	Disable http and only allow https.
	ODBC Report Access	Enabled with SSL encryption	
	Backup Health	Backup SAVED 2021-05-03T19:13:29+0000	
	Recent ssh activity	None	

Figure 2. Security checklist report following initial remedial actions.

Figure 3 shows the Security Checklist report after the administrator has reset the system's SSL Requirement setting to *HTTPS only*. All web browser communications are now encrypted.

The single remaining asterisk indicates that a more secure certificate should be used to improve your level of security. It is recommended that you use either a trusted certificate provided by a CA, or your organization's own certificate and matching key.

	Property 	Status	Recommendations
	Admin Account Password Age	Last updated 05/03/2021	
	Strong User Account Passwords	Minimum password length is set to 14	
*	SSL Certificate (for https)	Self-signed certificate expires Aug 6 20:23:18 2023 GMT	
	ODBC Report Access	Enabled with SSL encryption	
	Backup Health	Backup SAVED 2021-05-03T19:13:29+0000	
	Recent ssh activity	None	

Figure 3. Security checklist report indicating no serious security issues.

Appendix D: LenelS2 Node Hardening

To harden your LenelS2 nodes, we recommend that you:

- Require SSL for all connections, as described in the following section.
- Enable secure node communications ([page 43](#)).

For complete information on setting up LenelS2 nodes, see the *NetBox Initial Software Setup Guide* (#NB-SS-*nn*, where *nn* refers to the latest revision).

Requiring SSL Connections

When logging into a LenelS2 node for the first time to change the default password, we recommend that you select the **Require SSL** option to require that all connections to the node are secured with SSL.

To log into the LenelS2 node and change the default password:

1. Browse to the default IP address for the node (192.168.0.251) or to the address assigned to the node when it was set up.
2. Enter the password **admin** and click **Log in** to log into the LenelS2 node.
3. Click **Web Server** in the header bar to display the following page:

The screenshot shows a web interface for a LenelS2 node. At the top is a 'Change Password' section with three input fields: 'Current', 'New', and 'Re-enter', followed by a 'Change' button. Below this is a 'Security Policies' section containing three toggle switches: 'Allow logins when locked' (unchecked), 'Disable this web service when SSH is disabled' (unchecked), and 'Require SSL' (checked). An 'Update' button is at the bottom of the 'Security Policies' section.

4. Under Change Password, enter the current login password (*admin*), enter and re-enter a new, complex password, and click **Change**.

To create a complex password, follow the recommendations in [Guidelines for Creating Strong Passwords](#) on page 16.

5. Under Security Policies, select any or all of the following check boxes and click **Update**:
 - **Allow logins when locked**: Allows users to log into the LenelS2 node when the enclosure door is closed. We recommend leaving this option unselected.
 - **Disable this web service when SSL is disabled**: Ensures that the web server will be inaccessible to users when SSL is disabled.
 - **Require SSL** (Recommended): Requires that all connections to the node are secured with SSL. When this option is enabled, all `http://` requests are redirected to `https://`. The node will present a self-signed certificate, which the user will need to accept.

If Require SSL is not enabled and a person enters `https://` in a browser, the node will present a self-signed certificate, which the user will need to accept.

Note:

The login policy for LenelS2 nodes specifies that after 5 failed login attempts, the next attempt will be ignored and the user must wait for 4 minutes before trying again.

Enabling Secure Node Communications

Enabling secure communications for the LenelS2 node will allow it to exchange data securely with the Network Controller. Data sent between the node and controller will be protected through the use of a digital certificate installed on both.

Cryptographic key pairs provided by the digital certificate will be used to encrypt and sign the data, making it unreadable by unauthorized parties and verifying that your organization is the source of the data.

When you log into a LenelS2 node to enable secure communications, you will need to select the digital certificate that is currently configured for secure LenelS2 node communications on the Network Controller. This will be one of the following;

- **The default certificate**. This is a self-signed certificate that is installed on the Network Controller with software release 4.9.00 or later and automatically downloaded to LenelS2 nodes.

- **A custom certificate.** If a custom certificate was configured on the Network Controller and downloaded to LenelS2 nodes, it will be available for selection. For information on configuring a digital certificate on the Network Controller and downloading it to LenelS2 nodes, search for *Enabling Secure LenelS2 Node Communications* in the NetBox online help.
- **The LenelS2 Signed certificate.** This certificate is signed by LenelS2. It is installed on the Network Controller with software release 4.9.12 or later and automatically downloaded to LenelS2 nodes.

The LenelS2 Signed certificate is available only for nodes connected to NetBox Online. For all other nodes, we recommend using a custom certificate or, at a minimum, the default certificate.

To enable secure LenelS2 node communications:

1. Ensure that secure node communications is enabled on the Network Controller.
For instructions, search for *Enabling Secure LenelS2 Node Communications* in the NetBox online help.
2. Log into the LenelS2 node.
3. Click **Network** in the header bar.
4. Under Node Secure Communications, select **Enable**.
5. To use the default certificate or the LenelS2 Signed certificate, select it from the Certificate drop-down list and click **Submit**.
If you select the LenelS2 Signed certificate, be sure it is the certificate currently configured for secure node communications on the Network Controller.
6. To use a custom certificate:
 - a. For Certificate Type, select **Custom**.
 - b. Click **Choose File**.
 - c. Browse to and open the custom certificate file.
The certificate's file name and common name appear on the page. Be sure it is the certificate currently configured for secure node communications on the Network Controller.
 - d. Click **Submit**.
You should see a message indicating that the custom certificate has been successfully updated.

7. To confirm that the node has restarted and is connected to the Network Controller, log into the NetBox web interface and select **Configuration : Site Settings : Node Status**.

On the Node Status page, the LenelS2 node's UID will be displayed in green and it will have the status *Connected*. This should take just a few seconds.

Note:

Following a hardware reset of the node to factory defaults, you will need to re-enable secure communications for the node. If you use a custom certificate, it will need to be downloaded to the node again. Only the default certificate is retained on the node during a hardware reset to factory defaults.

If there is a secure node configuration error, a message in the Activity Log will indicate the reason for the error, as described in Table 4.

Table 4. Activity Log Messages for Secure Node Configuration Errors

Error Message	Meaning
secure connection attempt for <node>	Secure node communications is disabled on the Network Controller but enabled on the specified node.
certificate not found for <node>	The specified node is using an unknown digital certificate.
non-secure connection attempt for <node>	Secure node communications is enabled on the Network Controller but disabled on the specified node.

Appendix E: Acronyms Used in this Guide

A

- ACL: access-control list

C

- CA: certificate authority, or certification authority
- CIS: Center of Internet Security

D

- DoS: Denial of Service (cyberattack)
- DMZ: Demilitarized Zone (perimeter network, or screened subnet)

F

- FQDN: Fully Qualified Domain Name (for a specific host or server)
- FTP: File Transfer Protocol

H

- HSM: hardware security module
- HTTP: Hyper Text Transfer Protocol
- HTTPS: Hyper Text Transfer Protocol Secure

I

- IP: Internet Protocol
- ISO: International Organization for Standardization
- IT: Information Technology

L

- LDAP: Lightweight Directory Access Protocol
- LDAPS: LDAP over SSL

M

- MEM: Memory Exploit Mitigation
- MVP: Magic Video Player (LenelS2)
- MSP: Mobile Security Professional (LenelS2)
- MSU: Mobile Security User (LenelS2)

N

- NAS: Network Attached Storage
- NBAPI: Web-Based API for NetBox and NetBox Global
- NIST: National Institute of Standards and Technology

O

- OSDP: Open Supervised Device Protocol (access control)
- OWASP: Open Web Application Security Project

P

- PII: Personally identifiable information
- PoLP: Principle of least privilege

S

- SANS: SysAdmin, Audit, Network and Security Institute
- SMB: Secure Message Block (communication protocol)
- SSH: Secure Shell Protocol
- SSL: Secure Sockets Layer

T

- TCP: Transmission Control Protocol
- TLS: Transport Layer Security

U

- UDP: User Datagram Protocol

V

- VLAN: Virtual Local Area Network

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One Speen Street
Framingham, Massachusetts 01701 USA
Tel +1 508.663.2505
www.LenelS2.com